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BSC./B.A. SECOND SEMESTER EXAMINATIONS: 2022/2023

DEPARTMENT OF STATISTICS AND ACTUARIAL SCIENCE

ACTU 302: INTRODUCTION TO ACTUARIAL COMPUTING (3 CREDITS)

INSTRUCTIONS: ANSWER ALL QUESTIONS

- Create a folder on desktop and name it *ACTU302_2023_ID*. For example, a student with identity number (ID), 10070848, will name his/her folder: *ACTU302_2023_10070848*.
- For each question, you are required to provide your R codes and your results. Only a few results must be displayed as an illustration of your R codes to limit the number of pages for your work.
- Use Microsoft Word to organise your work and save it in BOTH MS Word and PDF format with your ID number as the file name. For example, a student with ID number 10070848 will have his/her files named: *ACTU302_10070848*. Upload these two files on SAKAI.
- Please adhere to these instructions to have your work properly submitted for marking

TIME ALLOWED: 2 $\frac{1}{2}$ HOURS

The **Insurance** data set in package **MASS** consists of the numbers of policyholders of an insurance company who were exposed to risk, and the numbers of car insurance claims made by those policyholders in the third quarter of 1973. Use this data set to answer the questions 1-3.

1. (a) What is the data structure and type of each variable? [4 marks]
(b) Select an appropriate descriptive statistics for each variable and comment on your results. [10 marks]
(c) Randomly select 50 observations from this data. Provide your R code and show the first few observations of your sampled data. [6 marks]
(d) Compute the correlation between the variables, **Claims** and **Holders**, to enable you comment on the extent of relationship. Test at 10% whether the relationship is significant. [10 marks]
(e) Plot the claims against the numbers of policyholders. Is the relationship supportive of your results in (d)? [6 marks]

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2. (a) Compute appropriate descriptive statistics of the claims per age category to aid in comparison of the claims of the various age groups. [5 marks]
 - (b) Does your descriptive statistics support the assertion of equality in the average claims? [2 marks]
 - (c) Conduct an appropriate statistical test at 5% significance level to check the equality of the average claims per age category. You must check the assumption of the selected test. [10 marks]
 - (d) From your results and at the same significance level of 5%, which age group has the highest claim amounts? [5 marks]
3. Consider a random variable X which is distributed as gamma with parameters α and β with probability density function

$$f(x) = \frac{\beta^\alpha}{\Gamma(\alpha)} x^{\alpha-1} e^{-\beta x}, \quad x > 0, \alpha, \beta > 0.$$

The moment and maximum likelihood estimators can be used to estimate the parameters α and β .

- (a) Use moment and maximum likelihood estimation methods to fit the gamma distribution to the Claims. [5 marks]
 - (b) Plot the relative frequency histogram of the claims data and superimpose the frequency curve of the two estimation methods. Which one provides the better fit? [10 marks]
4. Consider two investment projects shown in the table below and assume the risk free rate on the Bank of Ghana's T-bill is 19%.

Project A	-10,000	12,500	100	10	200	200
Project B	-10,000	-1000	0	100	200	20,000

- (a) With the help of an appropriate package in R, compute the net present value and internal rate of return for each investment Project. [10 marks]
- (b) Write your own function(s) to compute the Net Present Value. [10 marks]
- (c) Based on your results, which of the investment Project is more viable? Explain. [5 marks]